*.tem

Purpose

`.tem` is an older extension for temperature data files, used by some data sets and preprocessing tools. SWAT+ opens a temperature file by the name given in `tmp.cli` and does not look at the extension, so a `.tem` file is read in the same way as a `.tmp` file. SWAT+ Editor writes `.tmp`.

The name has to be the same, extension included, in two places:

- a line of `tmp.cli`, which tells SWAT+ to read the file, and
- the `tmp` column of `weather-sta.cli`, which assigns it to a weather station.

If the `tmp` column names a file that is not listed in `tmp.cli`, SWAT+ writes `<name> file not found (tgage)` to `diagnostics.out` and generates the temperatures of that station for the whole run.

Format

The format is that of a `.tmp` file: a title line, a header line, a gauge line with `nbyr`, `tstep`, `lat`, `lon` and `elev`, and one record per day with year, day of year, maximum and minimum temperature in °C. Values of -97 or less mark missing data.

Example

`refdata/Ames_sub1/ames.tem` in the SWAT+ repository:

```
AME.Tmp
 NBYR    TSTEP       LAT      LONG      ELEV
   121        0     42.04    -93.89    316
1900    1   -99 -99
1900    2   -99 -99
...
1975    1   4.4 -7.2
1975    2   5   -7.8
...
2020    365 -1.7    -7.8
2020    366 -2.8    -8.9
```

A daily record of 121 years, 1900 to 2020, separated by tabs, which free-format reading accepts. In this data set the `tmp` column of `weather-sta.cli` names `ames.tem`, while `tmp.cli` lists `ames.tmp`, which is an empty file. SWAT+ therefore does not find `ames.tem`, never reads it, and generates the temperatures of the Ames run. Listing `ames.tem` in `tmp.cli` instead of `ames.tmp` makes the run use the measured record.

Related files

- `*.tmp`: the same format; missing values, elevation adjustment and storage are described there.
- `tmp.cli`: lists the temperature files to read.
- `weather-sta.cli`: assigns a temperature file to each weather station.

Sources

- Reader: `cli_tmeas.f90`
- Type definition: `type climate_measured_data` in `climate_module.f90`

- Name lookup and the `file not found (tgage)` message: `cli_staread.f90`, `search.f90`
- Daily use: `climate_control.f90`